

Satisfiability of the Two-Variable Fragment of First-Order Logic with counting quantifiers over finite trees

(Spełnialność logiki z dwiema zmiennymi i kwantyfikatorami zliczającymi na
drzewach skończonych)

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Abstract

The finite satisfiability problem for the two-variable fragment of first-order logic interpreted over trees was recently shown to be EXPSPACE-complete. We show that adding counting quantifiers to the logic does not affect the complexity of the finite satisfiability problem.

W ostatnich latach wykazano, że problem skończonej spełnialności dla fragmentu logiki pierwszego rzędu z dwiema zmiennymi jest EXPSPACE-zupełny. W niniejszej pracy rozszerzymy tę logikę o kwantyfikatory zliczające i udowodnimy, że złożoność problemu skończonej spełnialności pozostaje taka sama.

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Chapter 1

Preliminaries and historical results

1.1 Short introduction

One of classical research results in computational logic is the undecidability of the satisfiability for first-order logic, proven by Church and Turing in the thirties of the last century. This result caused many questions concerning natural decidable fragments of first-order logic. Do they exist? If they do, how to find them? One of the most natural ideas during the search for these fragments is limiting the number of variables. It was revealed that a fragment that allows just three variables is already undecidable. Fortunately, if we limit ourselves to two variables, such logic has the property of a finite model and every satisfiable formula has a model of exponential size. This fact allows us to solve the satisfiability problem in NEXPTIME. Despite its limited expressive power, the logic obtained in this way has a lot of relations with query languages, descriptive logics, and it allows to embed modal logics. In this paper we will focus on two-variable logics interpreted over finite trees. This type of logics allows us to describe e.g. properties of databases - they are especially connected with popular XML format.

1.2 Preliminaries

1.2.1 Logics, trees and atomic types

We work with signatures of the form $\tau = \tau_0 \cup \tau_{nav}$, where τ_0 is a set of unary symbols and $\tau_{nav} = \{\downarrow, \downarrow^+, \rightarrow, \rightarrow^+\}$ is the set of *navigational* binary symbols. Over such signatures we consider two fragments of first-order logic: FO^2 , i.e., the restriction of first-order logic in which only variables x and y are available, and its extension with *counting quantifiers*, C^2 , in which quantifiers of the form $\exists^{\geq n}$, $\exists^{=n}$, $\exists^{\leq n}$, for $n \in \mathbb{N}$

are allowed. We assume that the reader is familiar with their standard semantics.

We write $\text{FO}^2[\tau_{bin}]$ or $\text{C}^2[\tau_{bin}]$ where $\tau_{bin} \subseteq \tau_{nav}$ to denote that the only binary symbols that are allowed in signatures are from τ_{bin} . We will mostly work with $\text{C}^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$, i.e., the fragment with counting quantifiers, but we will also use $\text{FO}^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$.

We are interested in finite unranked, ordered tree structures, in which the interpretation of the symbols from τ_{nav} is fixed: $\downarrow$ is interpreted as the child relation, $\rightarrow$ as the right sibling relation, and $\downarrow^+$ and $\rightarrow^+$ as their respective transitive closures. We read $u \downarrow w$ as " w is a *child* of u " and $u \rightarrow w$ as " w is the *right sibling* of u ". We will also use other standard terminology like *ancestor*, *descendant*, *preceding-sibling*, *following-sibling*, etc.

We use $x \not\sim y$ to abbreviate the formula stating that x and y are in *free position*, i.e., that they are related by none of the navigational binary predicates available in the signature. Let us call the formulas specifying the relative position of a pair of elements in a tree with respect to binary navigational predicates *order formulas*. There are ten possible order formulas: $x \downarrow y$, $y \downarrow x$, $x \downarrow^+ y \wedge \neg(x \downarrow y)$, $y \downarrow^+ x \wedge \neg(y \downarrow x)$, $x \rightarrow y$, $y \rightarrow x$, $x \rightarrow^+ y \wedge \neg(x \rightarrow y)$, $y \rightarrow^+ x \wedge \neg(y \rightarrow x)$, $x \not\sim y$, $x = y$. They are denoted, respectively, as: $\theta_{\downarrow}$, $\theta_{\uparrow}$, $\theta_{\downarrow\downarrow^+}$, $\theta_{\uparrow\uparrow^+}$, $\theta_{\rightarrow}$, $\theta_{\leftarrow}$, $\theta_{\rightarrow^+}$, $\theta_{\leftarrow^+}$, $\theta_{\not\sim}$, $\theta_{=}$. Let Θ be the set of these ten formulas.

We use symbol $\mathfrak{T}$ (possibly with sub- or superscripts) to denote tree structures. For a given tree $\mathfrak{T}$ we denote by T its universe. A *tree frame* is a tree over a signature containing no unary predicates. We will sometimes say that a tree frame $\mathfrak{T}_f$ is the tree frame *of* $\mathfrak{T}$, or that $\mathfrak{T}$ is *based* of $\mathfrak{T}_f$ if $\mathfrak{T}_f$ is obtained from $\mathfrak{T}$ by dropping the interpretation of all unary and common binary symbols. We say that a formula φ is satisfiable *over* a tree frame if it has a model based on this tree frame.

Given a tree $\mathfrak{T}$, we say that a node $v \in T$ is a *minimal* node (having some fixed property) if there is no $w \in T$ (having this property) such that $\mathfrak{T} \models w \downarrow^+ v$. A $\downarrow$ -path ($\rightarrow$ -path) is a sequence of nodes $v_1, \dots, v_k$ such that $\mathfrak{T} \models v_i \downarrow v_{i+1}$ ($\mathfrak{T} \models v_i \rightarrow v_{i+1}$), for $i = 1, \dots, k-1$. Given a $\downarrow$ -path ($\rightarrow$ -path) P we say that distinct nodes $v_1, \dots, v_l$ (having some fixed property) are *l smallest* elements (having this property) on P if for any other $v \in P$ (having this property) we have $\mathfrak{T} \models v_i \downarrow^+ v$ ($\mathfrak{T} \models v_i \rightarrow^+ v$) for $i = 1, \dots, l$. Analogously we define *maximal* and *biggest* elements.

An (atomic) *1-type* is a maximal satisfiable set of atoms or negated atoms with free variable x . Note that the numbers of atomic 1-types are bounded exponentially in the size of the signature. We often identify a type with the conjunction of all its elements. We denote by α_φ the set of 1-types over the signature consisting of symbols appearing in φ .

For a given τ -tree $\mathfrak{T}$, and a node $v \in T$ we say that v *realizes* a 1-type α if α is the unique 1-type such that $\mathfrak{T} \models \alpha[v]$. We denote by $\text{tp}^\mathfrak{T}(v)$ the 1-type realized by v .

1.2.2 Normal forms

As usual when working with satisfiability of two-variable logics we employ Scott-type normal form. We start with its adaptation for the case of $\text{FO}^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$.

Definition 1.2.1 ([2]). We say that an $\text{FO}^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ formula φ is in *normal form* if

$$\varphi = \forall xy \chi(x, y) \wedge \bigwedge_{i=1}^m \forall x (\lambda_i(x) \Rightarrow \exists y (\theta_i(x, y) \wedge \chi_i(x, y))),$$

where $\lambda_i(x)$ is an atomic formula $A(x)$ for some unary symbol A , $\chi(x, y)$ and $\chi_i(x, y)$ are quantifier-free, and $\theta_i(x, y)$ is an order formula.

For C^2 we use a similar but slightly different normal form then the one used in Definition 1.2.1. One obvious difference is that it uses counting quantifiers, the other is that its $\forall\exists$ -conjuncts do not need to contain the θ_i -components, specifying the position of the required witnesses. Refining the normal form by incorporating those components is possible but seems to require an exponential blow-up.

Definition 1.2.2. We say that a formula $\varphi \in \text{C}^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ is in *normal form*, if:

$$\varphi = \forall x \forall y \chi(x, y) \wedge \bigwedge_{i=1}^m (\forall x \exists^{\bowtie_i C_i} y \chi_i(x, y)),$$

where $\bowtie_i \in \{\leq, \geq\}$, each C_i is a natural number, and $\chi(x, y)$ and all $\chi_i(x, y)$ are quantifier-free.

The following lemma can be proved in a standard fashion.

Lemma 1.2.3 ([4]). Let φ be a formula from $\text{C}^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ over a signature τ . There exists a polynomially computable normal form $\text{C}^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ formula φ' over signature τ' consisting of τ and some additional unary symbols, such that φ and φ' are satisfiable over the same tree frames.

Given a normal form $\text{C}^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ formula φ and a tree $\mathfrak{T} \models \varphi$, we say that a node $w \in T$ is a *witness* for $v \in T$ and a conjunct $\forall x \exists^{\bowtie_i C_i} y \chi_i(x, y)$ of φ if $\mathfrak{T} \models \chi_i[v, w]$. If additionally $\mathfrak{T} \models w \downarrow^+ v$ then w is an *upper* witness, if $\mathfrak{T} \models v \downarrow^+ w$ then w is a *lower* witness, and so on. We also sometimes simply speak about $\rightarrow^+$ -witnesses, $\uparrow$ -witnesses, etc.

In Chapter 2, when a normal form formula φ is considered we always assume that it is as in Definition 1.2.2. In particular we allow ourselves, without explicitly recalling the shape of φ , to refer to its parameter m and components χ, χ_i .

1.2.3 Multisets

Any element of a model of a normal form conjunct $\forall x \exists^{\bowtie C} y \chi$ may require up to C witnesses, so we are interested in *multisets* counting these witnesses. To simulate

counting up to the value k , we use the function $cut_k : \mathbb{N} \rightarrow \{0, 1, 2, \dots, k, \infty\}$, where $cut_k(i) = i$ for $i \leq k$ and $cut_k(i) = \infty$ otherwise.

Formally, for a given $k \in \mathbb{N}$, a k -multiset M of elements from a set S is a function $M : S \rightarrow \{0, 1, 2, \dots, k, \infty\}$. For every element $e \in S$ we simply define $M(e)$, called the *multiplicity* of e in M , as the number of occurrences e in the multiset M , counted up to k . We employ standard set-theoretic operations, i.e., union $\cup$ and intersection $\cap$ with their natural semantics defined as follows: for given multisets A and B and an arbitrary element e from their domains, we define $(A \cup B)(e) = cut_k(A(e) + B(e))$ and $(A \cap B)(e) = \min(A(e), B(e))$. Additionally, we define the empty multiset $\emptyset$ as the function that for any argument returns 0 and the singleton $\{e\}$ of e as the function such that $\{e\}(e) = 1$ and $\{e\}(e') = 0$ for all $e' \neq e$.

1.2.4 Alternating Turing machines

An alternating Turing machine can be seen as a generalization of a the standard nondeterministic Turing machine and could be defined in the similar way. More formally:

Definition 1.2.4. An alternating Turing machine is a quintuple $M = \langle Q, \Gamma, \delta, q_0, g \rangle$, such that:

- $Q = \{q_0, q_1, \dots, q_{|Q|-1}\}$ is a finite set of states,
- $\Gamma = \{A_0, A_1, \dots, A_{|\Gamma|-1}\}$ is a finite tape alphabet,
- $\delta : Q \times \Gamma \rightarrow 2^{Q \times \Gamma \times \{L, R\}}$ is a transition function,
- $q_0 \in Q$ is an initial state, and
- $g : Q \rightarrow \{\forall, \exists, ACCEPT, REJECT\}$ specifies a type of each state.

The main difference between alternating and nondeterministic Turing machine is that if an alternating machine M is in a universal state $q_\forall$, then M accepts an input iff every transition from $q_\forall$ leads to an accepting state. Analogously, if M is in an existential state $q_\exists$, then M accepts an input iff some transition from $q_\exists$ leads to an accepting state. We will assume now that the reader is familiar with a concept of alternation and knows the standard semantics of these machines.

By $ATIME(f(n))$ we denote the set of problems solvable by an alternating Turing machine working in alternating time (bounded by the function) f . A well-known theorem states that:

Theorem 1.2.5 ([1]). $AEXPTIME = EXPSPACE$,

which will be heavily used in the next chapter.

1.2.5 Full types, witness counting and reduced types

Definition 1.2.6 (Full type). A k -full type $\bar{\alpha}$ (over a signature $\tau = \tau_0 \cup \tau_{nav}$) is a function $\bar{\alpha} : \Theta \rightarrow \{0, 1, 2, \dots, k, \infty\}^{2^{\tau_0}}$, i.e., a function which takes a position from Θ and returns a k -multiset of 1-types over τ , that satisfies the following conditions:

- $\bar{\alpha}(\theta_{\uparrow}), \bar{\alpha}(\theta_{\rightarrow}), \bar{\alpha}(\theta_{\leftarrow})$ is either empty or a singleton,
- $\bar{\alpha}(\theta_{=})$ is a singleton, and
- if $\bar{\alpha}(\theta_{\uparrow})$ (respectively, $\bar{\alpha}(\theta_{\downarrow}), \bar{\alpha}(\theta_{\rightarrow}), \bar{\alpha}(\theta_{\leftarrow})$) is empty, then also the multiset $\bar{\alpha}(\theta_{\uparrow+})$ (respectively, $\bar{\alpha}(\theta_{\downarrow+}), \bar{\alpha}(\theta_{\rightarrow+}), \bar{\alpha}(\theta_{\leftarrow+})$) is empty.

Let C be the function that for a given normal form φ returns $C(\varphi) = \max\{C_i\}_{1 \leq i \leq m}$. We work with k -full types usually in contexts in which a normal form φ is fixed, and we are then particularly interested in $C(\varphi)$ -full types. The purpose of a k -full type is to say for a given node v , for each $\theta \in \Theta$ and each 1-type α' , how many vertices (counting up to k) of 1-type α' are in position θ to v . Formally:

Definition 1.2.7. For a given tree $\mathfrak{T}$ and $v \in T$ we denote by $\text{ftp}_k^{\mathfrak{T}}(v)$ the unique k -full type realized by v , i.e., the k -full type $\bar{\alpha}$ such that $\bar{\alpha}(\theta_{=})$ contains the 1-type of v and for all positions $\theta \in \Theta$ and for all atomic 1-types α' we have that

$$\bar{\alpha}(\theta)(\alpha') = \text{cut}_k \left(\#\{w \in T : \mathfrak{T} \models \theta[v, w] \wedge \text{tp}^{\mathfrak{T}}(w) = \alpha'\} \right)$$

where $\#S$ denotes the cardinality of the set S .

We next define functions which for a normal form φ and a $C(\varphi)$ -full type $\bar{\alpha}$ say how many witnesses a realization of $\bar{\alpha}$ has for each of the conjuncts of φ in all possible positions θ .

Definition 1.2.8 (Witness counting functions). Let φ be a normal form formula, and let $\bar{\alpha}$ be a $C(\varphi)$ -full type. Assume that $\bar{\alpha}(\theta_{=}) = \{\alpha\}$. We associate with φ and $\bar{\alpha}$ a function $W_{\bar{\alpha}}^{\varphi} : \{1, \dots, m\} \times \Theta \rightarrow \{0, 1, \dots, C(\varphi), \infty\}$, whose values are defined in the following way:

- for $\theta \in \{\theta_{=}, \theta_{\rightarrow}, \theta_{\leftarrow}, \theta_{\downarrow}, \theta_{\uparrow}\}$ and any i :

$$W_{\bar{\alpha}}^{\varphi}(i, \theta) = \begin{cases} 1 & \text{if } \bar{\alpha}(\theta) = \{\alpha'\} \text{ and } \alpha(x) \wedge \alpha'(y) \wedge \theta(x, y) \models \chi_i(x, y) \\ 0 & \text{otherwise,} \end{cases}$$

- for $\theta \in \{\theta_{\leftarrow+}, \theta_{\rightarrow+}, \theta_{\downarrow+}, \theta_{\uparrow+}, \theta_{\nearrow}\}$ and any i :

$$W_{\bar{\alpha}}^{\varphi}(i, \theta) = \text{cut}_{C(\varphi)} \left(\sum_{\alpha' : \alpha(x) \wedge \alpha'(y) \wedge \theta(x, y) \models \chi_i(x, y)} (\bar{\alpha}(\theta))(\alpha') \right).$$

This way $W_{\bar{\alpha}}^{\varphi}(i, \theta)$ is the number of witnesses (counted up to $C(\varphi)$), in relative position θ , for a node of full type $\bar{\alpha}$ and the formula χ_i from φ .

Now we relate the notion of full types with the satisfaction of normal form formulas.

Definition 1.2.9 (φ -consistency). Let φ be a $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ formula in normal form. Let $\bar{\alpha}$ be a $C(\varphi)$ -full type. Assume that $\bar{\alpha}(\theta_{=})$ consists of a 1-type α . We say that $\bar{\alpha}$ is φ -consistent if it satisfies the following conditions.

- $\alpha(x) \models \chi(x, x)$,
- $\alpha(x) \wedge \alpha'(y) \wedge \theta(x, y) \models \chi(x, y)$ (for all $\theta \in \Theta$, $\alpha' \in \bar{\alpha}(\theta)$), and
- for all $1 \leq i \leq m$ the inequality $\sum_{\theta \in \Theta} W_{\bar{\alpha}}^{\varphi}(i, \theta) \bowtie_i C_i$ holds.

Lemma 1.2.10. Assume that a formula $\varphi \in C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ is in normal form. Then $\mathfrak{T} \models \varphi$ iff every $C(\varphi)$ -full type realized in $\mathfrak{T}$ is φ -consistent.

Proof. $\implies$

Assume that $\mathfrak{T} \models \varphi$. Let $\bar{\alpha}$ be a $C(\varphi)$ -full type realized in $\mathfrak{T}$. We have $\mathfrak{T} \models \forall x \forall y \chi(x, y)$, and $\mathfrak{T} \models \forall x \exists^{\bowtie_i C_i} y \chi_i(x, y)$ for all i . The first two conditions from Definition 1.2.9 are straightforward, because χ is true for every pair of vertices, every pair of vertices is related by some $\theta \in \Theta$ and every vertex has its own 1-type. For the third condition, take $v \in T$, such that $\text{ftp}_{C(\varphi)}^{\mathfrak{T}}(v) = \bar{\alpha}$, and $i \in \mathbb{N}$. The number of $w \in T$, such that $\mathfrak{T} \models \chi_i[v, w]$, is $\bowtie_i C_i$, because $\mathfrak{T} \models \forall x \exists^{\bowtie_i C_i} y \chi_i(x, y)$.

$\longleftarrow$

Every pair of vertices is related with some $\theta \in \Theta$. Let $\bar{\alpha}, \bar{\beta}$ be the $C(\varphi)$ -full types of nodes $v, w \in T$ realized in $\mathfrak{T}$. By assumption $\bar{\alpha}, \bar{\beta}$ are φ -consistent, which proves (by the first and the second condition from Definition 1.2.9) that $\mathfrak{T} \models \forall x \forall y \chi(x, y)$.

Fix a vertex $v \in T$, its $C(\varphi)$ -full type $\bar{\alpha} = \text{ftp}_{C(\varphi)}^{\mathfrak{T}}(v)$ and some $i \in \mathbb{N}$. We know that $\bar{\alpha}$ is φ -consistent, so $\sum_{\theta \in \Theta} W_{\bar{\alpha}}^{\varphi}(i, \theta) \bowtie_i C_i$. By this fact, $\#\{w \in T \mid \exists \theta \in \Theta \theta(w, v)\} \bowtie_i C_i$, which means that we have the right number of witnesses for v to satisfy the formula χ_i . That gives us $\mathfrak{T} \models \forall x \exists^{\bowtie_i C_i} y \chi_i(x, y)$. We have shown that every conjunct from φ is true in $\mathfrak{T}$, so $\mathfrak{T} \models \varphi$. $\square$

The next notion will be used to describe information from full types in a (lossy) compressed form. We need this form to obtain tight complexity bounds.

Definition 1.2.11 (φ -reduced type). Let φ be a normal form $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ formula. For a given $C(\varphi)$ -full type $\bar{\alpha}$, its φ -reduced form, $\text{rftp}_{\varphi}(\bar{\alpha})$, is the tuple $(\alpha, W_{\bar{\alpha}}^{\varphi}, A, B, F)$, where $A = \bar{\alpha}(\theta_{\uparrow}) \cup \bar{\alpha}(\theta_{\uparrow\uparrow+})$, $B = \bar{\alpha}(\theta_{\downarrow}) \cup \bar{\alpha}(\theta_{\downarrow\downarrow+})$, $F = \bar{\alpha}(\theta_{\rightarrow}) \cup \bar{\alpha}(\theta_{\leftarrow}) \cup \bar{\alpha}(\theta_{\rightarrow+}) \cup \bar{\alpha}(\theta_{\leftarrow+}) \cup \bar{\alpha}(\theta_{\neq+}) \cup \bar{\alpha}(\theta_{\neq})$ and $\bar{\alpha}(\theta_{=})$ is the singleton of the 1-type α . If the $C(\varphi)$ -full type $\bar{\alpha}$ is realized by a vertex v in $\mathfrak{T}$ then we say that $\text{rftp}_{\varphi}(\bar{\alpha})$ is the φ -reduced type of v . This reduced full type will be denoted also as $\text{rftp}_{\varphi}^{\mathfrak{T}}(v)$.

Intuitively, if a k -full type $\bar{\alpha}$ is realized by a vertex v in a structure $\mathfrak{T}$ then the multisets A, B, F in $\text{rftp}_\varphi(\bar{\alpha})$ are respectively the k -multisets of 1-types realized in $\mathfrak{T}$ above, below and in free position to v .

Let $\bar{\alpha}, \bar{\beta}$ be k -full types. A *combined k -full type* is a k -full type $\bar{\gamma}$, such that $\bar{\gamma}(\theta) = \bar{\alpha}(\theta)$ or $\bar{\gamma}(\theta) = \bar{\beta}(\theta)$ for all positions $\theta \in \Theta$.

Lemma 1.2.12. Let $\bar{\alpha}, \bar{\beta}$ be φ -consistent $C(\varphi)$ -full types such that their φ -reduced forms are equal. Then the combined $C(\varphi)$ -full type $\bar{\gamma}$ in form $\bar{\gamma}(\theta) = \bar{\alpha}(\theta)$ for $\theta \in \{\theta_\uparrow, \theta_{\uparrow\uparrow+}, \theta_{\rightarrow}, \theta_{\rightarrow+}, \theta_{\nearrow}, \theta_{\leftarrow}, \theta_{\leftarrow+}\}$ and $\bar{\gamma}(\theta) = \bar{\beta}(\theta)$ for $\theta \in \{\theta_{\downarrow}, \theta_{\downarrow\downarrow+}\}$ is also φ -consistent.

Proof. Obviously $\bar{\gamma}$ satisfies the first two conditions from Definition 1.2.9 because $\bar{\alpha}$ and $\bar{\beta}$ do. The third condition is guaranteed by the equality of the witness counting components. $\square$

Example. Note that the assumption about equality of φ -reduced full types, and in particular their witness counting components, is essential. In [2, Proposition 2] the authors prove that in the setting without counting quantifiers a combined type remains φ -consistent without the assumption about equality of the witness-counting components. The following example shows that in our scenario it is no longer true.

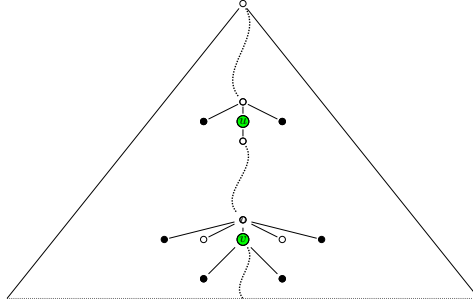


Figure 1.1: Naive combination of full types

Let φ be a formula saying that every green vertex has at most three direct black neighbors below, on the left or on the right; formally

$$\varphi = \forall x \exists^{\leq 3} y \text{ green}(x) \Rightarrow (\text{black}(y) \wedge (x \downarrow y \vee y \downarrow x \vee x \rightarrow y \vee y \rightarrow x)).$$

Let $\mathfrak{T}$ be a tree model from Fig. 1.2.5. Denote $\bar{\alpha} = \text{ftp}_{C(\varphi)}^{\mathfrak{T}}(u)$ and $\bar{\beta} = \text{ftp}_{C(\varphi)}^{\mathfrak{T}}(v)$. Because $\mathfrak{T} \models \varphi$, the $C(\varphi)$ -full types $\bar{\alpha}$ and $\bar{\beta}$ are φ -consistent. However the combined $C(\varphi)$ -full type $\bar{\gamma}$, in form described in Lemma 1.2.12, is not φ -consistent (the black nodes appear in $\bar{\gamma}$ on positions $\theta_{\downarrow}, \theta_{\leftarrow}, \theta_{\rightarrow}$ four times in total).

Chapter 2

$C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ on trees

In this chapter we prove that the finite satisfiability problem for $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ over trees is EXPSpace-complete. Intuitively, the proof is a combination of the two proofs from [2] and [3] that solve the problem for FO^2 on trees and for C^2 on linear orders respectively (note that a linear order is just a tree whose each node has at most one child). However, the method in [2] heavily depends on the normal form from Definition 1.2.1 where each conjunct corresponds to at most one relative position $\theta \in \Theta$. Although it is possible to bring a $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ formula into an analogous normal form, it requires an exponential blowup (dividing a set of witnesses into 10 subsets corresponding to 10 order formulas can be done in exponentially many ways). Therefore, to keep the complexity under control, we stay with usual, less refined normal form from Definition 1.2.2, but to compensate it we introduce a novel technique combining type information with witness counting.

2.1 The lower bound

Beside the fact that the lower bound on finite satisfiability of $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ comes from [2], where the EXPSpace-completeness of $FO^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ has been proved, it is possible to provide an alternative, independent proof based on encoding computations of AEXPTIME-Turing machine inside our models. It can be done by following general scheme of AEXPTIME-reduction presented in [5]. Because such encoding is quite technical, we will only show a first step, namely enforcing models to have doubly-exponential size.

Lemma 2.1.1. There exists a satisfiable formula $\varphi \in C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ such every its model has doubly-exponential size in $|\varphi|$.

Proof. Assume that the unary symbols $V_0, V_1, \dots, V_{n-1}, H_0, H_1, \dots, H_{n-1}$ are allowed in the signature. We will use these symbols to encode in binary two numbers - Pos_V and Pos_H . For a fixed vertex v in a tree, we calculate the value of $Pos_V(v)$ and $Pos_H(v)$ in the following way: $Pos_V(v) = \sum_{i: V_i(v)} 2^i$ and $Pos_H(v) = \sum_{i: H_i(v)} 2^i$.

Let us sketch our approach. As a first step, we will write some obvious facts about the desired tree. Next, we will make use of the unary predicates V_i and the number encoded in vertical position to enforce that every $\downarrow$ -path starting from root contains each natural number from 0 to $2^n - 1$ as a vertical position. Finally, we guarantee that every non-leaf vertex has exactly 2^n children - it could be done in the similar way, namely using $\rightarrow$ -paths and unary predicates H_i .

All of the desired properties of such encoding:

1. There exists a root and $Pos_V(\text{root}) = 0$,
2. $Pos_V(\text{leaf}) = 2^n - 1$ for all leaves,
3. every non-leaf vertex has a leftmost child with horizontal number 0,
4. every non-leaf vertex has a rightmost child with horizontal number $2^n - 1$,
5. $\forall v \in V \forall w \in V \ v \downarrow w \implies Pos_V(w) = Pos_V(v) + 1$, and
6. $\forall v \in V \forall w \in V \ v \rightarrow w \implies Pos_H(w) = Pos_H(v) + 1$.

We denote by $\varphi_1, \varphi_2, \varphi_3, \varphi_4, \varphi_5, \varphi_6$ the sentences describing the above properties. It's quite easy to write the first four of them:

$$\varphi_1 = \exists x \underbrace{(\exists^{=0} y \ y \downarrow x)}_{\text{No vertices above}} \wedge \underbrace{\left(\bigwedge_{i=0}^{n-1} \neg V_i(x) \right)}_{Pos_V(x)=0}$$

$$\varphi_2 = \forall x \left(\underbrace{\left(\bigwedge_{i=0}^{n-1} V_i(x) \right)}_{Pos_V(x)=2^n-1} \iff \underbrace{(\exists^{=0} y \ x \downarrow y)}_{\text{No vertices below}} \right)$$

$$\varphi_3 = \forall x \left(\underbrace{\neg \left(\bigwedge_{i=0}^{n-1} V_i(x) \right)}_{x \text{ is not a leaf}} \implies \left(\underbrace{\exists y \ x \downarrow y}_{\text{has a son } y} \wedge \underbrace{\exists^{=0} x \ x \rightarrow y}_{y \text{ is leftmost}} \wedge \underbrace{\left(\bigwedge_{i=0}^{n-1} \neg H_i(y) \right)}_{Pos_H(y)=0} \right) \right)$$

$$\varphi_4 = \forall x \left(\underbrace{\neg \left(\bigwedge_{i=0}^{n-1} V_i(x) \right)}_{x \text{ is not a leaf}} \Rightarrow \left(\underbrace{\exists y \ x \downarrow y}_{\text{has a son } y} \wedge \underbrace{\exists^{=0} x \ y \rightarrow x}_{y \text{ is rightmost}} \wedge \underbrace{\left(\bigwedge_{i=0}^{n-1} H_i(y) \right)}_{Pos_H(y)=2^n-1} \right) \right)$$

Now we are going to formalize the fifth and sixth property from the above list. To do it, we need to write a formula $+1_V(x, y)$, which states that $Pos_V(y) = Pos_V(x) + 1$, and analogous formula $+1_H(x, y)$ for the horizontal purposes.

The original idea is very simple. First we look for the least significant zero bit in the binary representation of $Pos_V(x)$. If we find one (let it be the i -th bit), we reset all the previous bits in $Pos_V(y)$, we set its value to one, and rewrite all other bits from $Pos_V(x)$ to $Pos_V(y)$. Thus the formula $+1_V(x, y)$ could be written in the following way:

$$\bigwedge_{i=0}^{n-1} \left(\underbrace{\left(\neg V_i(x) \wedge \bigwedge_{j=0}^{i-1} V_j(x) \right)}_{\text{Look for the first zero bit}} \Rightarrow \left(\underbrace{\bigwedge_{j=0}^{i-1} \neg V_j(y)}_{\text{Reset first bits}} \wedge V_i(y) \wedge \underbrace{\bigwedge_{j=i+1}^{n-1} (V_j(x) \Leftrightarrow V_j(y))}_{\text{Rewrite last bits}} \right) \right)$$

Analogously we can write the formula $+1_H(x, y)$ as:

$$\bigwedge_{i=0}^{n-1} \left(\underbrace{\left(\neg H_i(x) \wedge \bigwedge_{j=0}^{i-1} H_j(x) \right)}_{\text{Look for the first zero bit}} \Rightarrow \left(\underbrace{\bigwedge_{j=0}^{i-1} \neg H_j(y)}_{\text{Reset first bits}} \wedge H_i(y) \wedge \underbrace{\bigwedge_{j=i+1}^{n-1} (H_j(x) \Leftrightarrow H_j(y))}_{\text{Rewrite last bits}} \right) \right)$$

Using the above formulas we can easily write the formula φ_5 and φ_6 as:

$$\varphi_5 = \forall x \forall y \ (x \downarrow y \implies +1_V(x, y))$$

$$\varphi_6 = \forall x \forall y \ (x \rightarrow y \implies +1_H(x, y))$$

Now we are ready to finish the proof of the lemma. Let $\varphi = \bigwedge_{i=1}^6 \varphi_i$. Observe that the length of φ is a polynomial function of the value n . On the other hand, the number of nodes in any tree model $\mathfrak{T} \models \varphi$ is equal to $\sum_{i=0}^{2^n-1} (2^n)^i = 2^{2^n \cdot n} (2^n + 1)$, which clearly is doubly-exponential in the size of n and finishes the proof. $\square$

2.2 The upper bound

In this section, we will focus on proving EXPSPACE-upper bound for $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$.

2.2.1 Small model theorem

We will demonstrate the small-model property of the logic, showing that every satisfiable formula φ has a tree model of depth and degree bounded exponentially in $|\varphi|$. It is obtained by first shortening $\downarrow$ -paths and then shortening the $\rightarrow$ -paths.

Theorem 2.2.1 (Small model theorem). Let φ be a formula of $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ in normal form. If φ is satisfiable then it has a tree model in which every path has length bounded by $3 \cdot (C(\varphi) + 2)^{10m+1} \cdot |\alpha_\varphi|^2$ and every vertex has degree bounded by $(4C(\varphi)^2 + 8C(\varphi)) \cdot |\alpha_\varphi|^5$.

We split the proof of this theorem into two parts. In Section 2.2.1 we show how to reduce the length of paths in a tree and in Section 2.2.2 we show how to reduce the degree of every vertex.

Short paths

Lemma 2.2.2 (Cutting lemma). Let $\varphi \in C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ be a formula in normal form and $\mathfrak{T}$ be its tree model. If there are two vertices $u, v \in T$, such that v is below u and $\text{rftp}_\varphi^\mathfrak{T}(u) = \text{rftp}_\varphi^\mathfrak{T}(v)$, then the tree $\mathfrak{T}'$, obtained by replacing the subtree rooted at u by the subtree rooted at v , is also a model of φ .

Proof. The proof goes by case analysis. First, observe that the $C(\varphi)$ -full type of u in tree $\mathfrak{T}'$ is a combination of the $C(\varphi)$ -full types of u and v in $\mathfrak{T}$ and thus, by Lemma 1.2.12, it is φ -consistent. In the rest of the proof we show that for every vertex w in $\mathfrak{T}'$ we have $\text{ftp}_{C(\varphi)}^\mathfrak{T}(w) = \text{ftp}_{C(\varphi)}^{\mathfrak{T}'}(w)$. Then Lemma 1.2.10 guarantees that the obtained tree $\mathfrak{T}'$ is indeed a model of φ .

Let w be any vertex from T' , or equivalently, a vertex from T such that w lies inside the tree rooted at v or lies outside of the tree rooted at u . There are four possible "locations" for w :

- (a) w is above u (cf. Fig. 2.1)
- (b) w is below v (cf. Fig. 2.2)
- (c) w is in a free position to u (cf. Fig. 2.3)
- (d) w is a sibling of u

For the rest of the proof, denote $\bar{\alpha} = \text{ftp}_{C(\varphi)}^\mathfrak{T}(w)$ and $\bar{\beta} = \text{ftp}_{C(\varphi)}^{\mathfrak{T}'}(w)$. Let us consider the four cases distinguished above.

(a) w is above u

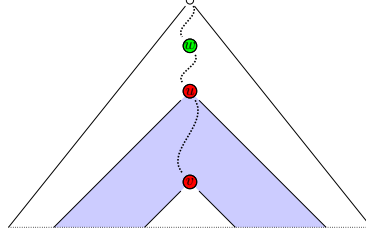


Figure 2.1: Case (a)

Of course cutting vertices between u and v does not change anything above or in free position to w , so obviously for all $\theta \in \{\theta_-, \theta_+, \theta_{\uparrow\uparrow+}, \theta_{\leftarrow}, \theta_{\leftarrow+}, \theta_{\rightarrow}, \theta_{\rightarrow+}, \theta_{\nearrow}\}$ we have $\bar{\alpha}(\theta) = \bar{\beta}(\theta)$. The 1-type of the node immediately below w is also not changed, so $\bar{\alpha}(\theta_{\downarrow}) = \bar{\beta}(\theta_{\downarrow})$. The only missing case is equality of multisets $\bar{\alpha}(\theta_{\downarrow\downarrow+})$ and $\bar{\beta}(\theta_{\downarrow\downarrow+})$, but it follows from the equality of the φ -reduced types $\text{rftp}_{\varphi}^{\mathfrak{T}}(u)$ and $\text{rftp}_{\varphi}^{\mathfrak{T}}(v)$, and in particular their B -components. More specifically, for a given 1-type γ , the number of occurrences of γ in $\bar{\alpha}(\theta_{\downarrow\downarrow+})$ is (counted up to $C(\varphi)$) the number of occurrences of γ in the subtree of $\mathfrak{T}$ rooted at w . We can divide this tree into three pieces, as in Fig. 2.1: the upper part without the subtree rooted at u , the lower part rooted at v and the remaining middle part. Now, the multiplicity of γ in $\bar{\alpha}(\theta_{\downarrow\downarrow+})$ is simply the sum of multiplicities of γ in each of these parts. But from the fact that $\text{rftp}_{\varphi}^{\mathfrak{T}}(u) = \text{rftp}_{\varphi}^{\mathfrak{T}}(v)$, we know that the multiplicity of γ in the subtree rooted at u is the same as in the subtree rooted at v . It means that either there are more than $C(\varphi)$ occurrences of γ in the subtree rooted at v or there are no occurrences of γ in the middle part. In both cases the multiplicity of γ below w is the same before and after the surgery.

(b) w is below v

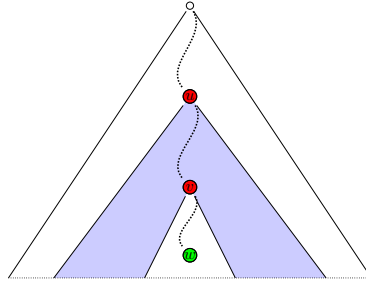


Figure 2.2: Case (b)

The reasoning to show that $\bar{\alpha}(\theta) = \bar{\beta}(\theta)$ for $\theta \in \{\theta_-, \theta_+, \theta_{\uparrow\uparrow+}, \theta_{\leftarrow}, \theta_{\leftarrow+}, \theta_{\rightarrow}, \theta_{\rightarrow+}, \theta_{\downarrow\downarrow+}\}$ is the same or symmetric to the previous case.

It is a bit more tricky to prove that $\bar{\alpha}(\theta_{\nearrow}) = \bar{\beta}(\theta_{\nearrow})$. Consider an arbitrary 1-type γ . Observe that vertices of type γ in free position to w in $\mathfrak{T}$ are vertices of type γ that are

- in free position to w in the subtree rooted at v (denote the number of them by $\#w_{\nearrow}$),
- close and distant siblings of v (denoted $\#v_{\rightleftharpoons}$),
- vertices in free position to v in the subtree rooted at u (denoted $\#v_{\nearrow}$),
- close and distant siblings of u (denoted $\#u_{\rightleftharpoons}$), or
- in free position to u in the whole tree ($\#u_{\nearrow}$).

This gives us the equation:

$$(\bar{\alpha}(\theta_{\nearrow}))(\gamma) = \text{cut}_{C(\varphi)}(\#w_{\nearrow} + \#v_{\rightleftharpoons} + \#v_{\nearrow} + \#u_{\rightleftharpoons} + \#u_{\nearrow}).$$

By the assumption that $\text{rftp}_{\varphi}^{\mathfrak{T}}(v) = \text{rftp}_{\varphi}^{\mathfrak{T}}(u)$, the F -components of these φ -reduced $C(\varphi)$ -full types are equal, so $\text{cut}_{C(\varphi)}(\#v_{\rightleftharpoons} + \#v_{\nearrow} + \#u_{\rightleftharpoons} + \#u_{\nearrow}) = \text{cut}_{C(\varphi)}(\#v_{\rightleftharpoons} + \#v_{\nearrow})$. It means that either $\text{cut}_{C(\varphi)}(\#u_{\rightleftharpoons} + \#u_{\nearrow}) = \infty$ or $\#v_{\rightleftharpoons} + \#v_{\nearrow} = 0$. In both cases, $\bar{\alpha}(\theta_{\nearrow})$ does not change after removing vertices between nodes u and v .

(c) w is in a free position to u

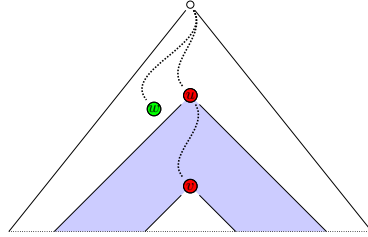


Figure 2.3: Case (c)

The fact that $\bar{\alpha}(\theta) = \bar{\beta}(\theta)$ for $\theta \in \{\theta_{\downarrow}, \theta_{\downarrow+}, \theta_{\leftarrow}, \theta_{\leftarrow+}, \theta_{\rightarrow}, \theta_{\rightarrow+}, \theta_{\uparrow}, \theta_{\uparrow+}, \theta_{=}\}$ is quite obvious (for the same reason as in the previous cases).

We will show that $\bar{\alpha}(\theta_{\nearrow}) = \bar{\beta}(\theta_{\nearrow})$. Observe that the whole tree rooted at u is in free position to w . $C(\varphi)$ -full types of all vertices w' outside this tree, such that $\mathfrak{T} \models \theta_{\nearrow}[w, w']$ don't change, so we can concentrate only on vertices from that tree. Note that $\text{rftp}_{\varphi}^{\mathfrak{T}}(u) = \text{rftp}_{\varphi}^{\mathfrak{T}}(v)$, which means equality of "below" multisets. Because multiplicity of each 1-type γ below v is equal to the multiplicity of γ below u , we have equal multiplicity of vertices in free position to w before and after replacing the root.

(d) w is a sibling of u

The proof is similar to the previous ones. We only need to show that $\bar{\alpha}(\theta_{\nearrow}) = \bar{\beta}(\theta_{\nearrow})$. Observe that $\text{ftp}_{C(\varphi)}^{\mathfrak{T}}(u)(\theta_{\downarrow}) \cup \text{ftp}_{C(\varphi)}^{\mathfrak{T}}(u)(\theta_{\downarrow+}) = \text{ftp}_{C(\varphi)}^{\mathfrak{T}}(v)(\theta_{\downarrow}) \cup$

$\text{ftp}_{C(\varphi)}^{\mathfrak{T}}(v)(\theta_{\downarrow+})$, so we don't accidentally cut any of the free witnesses of w , which proves the desired equality. $\square$

Lemma 2.2.3. Let φ be a satisfiable formula in normal form of $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ satisfied in a finite tree. Then there exists a tree model of φ whose every $\downarrow$ -path has length bounded by $3 \cdot (C(\varphi) + 2)^{10m+1} \cdot |\alpha_\varphi|^2$.

Proof. According to Lemma 2.2.2 we can restrict attention to models with the property that every φ -reduced full type appears only once on every $\downarrow$ -path. Let $\mathfrak{T} \models \varphi$ be a tree model with this property. Let $v_1, v_2, \dots, v_n$ be a $\downarrow$ -path in $\mathfrak{T}$. Observe that the φ -reduced full types on this path behave in a monotonic way in the sense that for every i and the φ -reduced full types of the $i, (i+1)$ -th vertices $R_i = (\alpha_i, W_i, A_i, B_i, F_i)$ and $R_{i+1} = (\alpha_{i+1}, W_{i+1}, A_{i+1}, B_{i+1}, F_{i+1})$, we have $A_i \subseteq A_{i+1}, B_{i+1} \subseteq B_i$ and $F_i \subseteq F_{i+1}$. A 1-type α can occur in a multiset from 0 to $C(\varphi)$ times. If α appears more than $C(\varphi)$ times, its multiplicity is ∞ . Hence the number of modifications of each multiset from A, B, F is bounded by $(C(\varphi) + 2) \cdot |\alpha_\varphi|$. There are up to $|\alpha_\varphi| \cdot (C(\varphi) + 2)^{10m}$ φ -reduced full types with fixed multisets A, B, F (because it is the number of all possible 1-types multiplied by the number of all possible witness-counting functions). Combination of these two observations gives us the desired estimation $(C(\varphi) + 2)^{10m+1} \cdot |\alpha_\varphi|^2 \cdot 3$. $\square$

Small degree

Definition 2.2.4. For a given vertex $v \in T$ and its $C(\varphi)$ -full type $\text{ftp}_{C(\varphi)}^{\mathfrak{T}}(v) = \bar{\alpha}$, the *horizontal $C(\varphi)$ -full type* of v in $\mathfrak{T}$ is the quintuple

$$\text{hftp}_{C(\varphi)}^{\mathfrak{T}}(v) = (\bar{\alpha}(\theta_{=}), \bar{\alpha}(\theta_{\rightarrow}), \bar{\alpha}(\theta_{\rightarrow+}), \bar{\alpha}(\theta_{\leftarrow}), \bar{\alpha}(\theta_{\leftarrow+})).$$

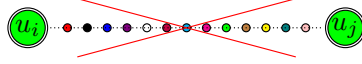
Lemma 2.2.5. Let φ be a formula in normal form of $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ satisfied in a finite tree $\mathfrak{T}$. Then there exists a tree model of φ , obtained by removing some subtrees from $\mathfrak{T}$, such that the degree of every vertex is bounded by $(4C(\varphi)^2 + 8C(\varphi)) \cdot |\alpha_\varphi|^5$.

Proof. First, we will show how to limit the degree of a single vertex. After that we can traverse the tree in depth-first manner and, by cutting unwanted vertices, obtain the desired model. Let v be a vertex from T . We denote by $\text{Children}(v)$ the set $\{u_1, u_2, \dots, u_N\}$ of all children of v (ordered by $\rightarrow_+$). For every 1-type α , we are going to mark some elements of $\text{Children}(v)$ and, after that, limit number of vertices between two adjacent marked ones.

Let $U_\alpha = \{u_i \mid \text{tp}^{\mathfrak{T}}(u_i) = \alpha\}$ be the set of children of v with the 1-type α and let $U_\alpha^\downarrow = \{u_i \mid \exists w \in T (u_i \downarrow w \vee u_i \downarrow \downarrow^+ w) \wedge \text{tp}^{\mathfrak{T}}(w) = \alpha\}$ be the set of children of v with descendants of 1-type α . For every 1-type α , we mark $\min(C(\varphi), |U_\alpha|)$ vertices from

U_α and $\min(C(\varphi), |U_\alpha^\downarrow|)$ vertices from $U_\alpha^\downarrow$ (it is important to mark u_1 and u_N during this process). Marked vertices are the required witnesses for vertex v . The number of marked vertices ensures us that during the cutting we do not lose also any of free witnesses for any other vertex in the tree $\mathfrak{T}$. It's easy to see that during this process we marked at most $2C(\varphi) \cdot |\alpha_\varphi|$ vertices.

Now the reasoning is similar to that of Lemma 2.2.2. Let u_i, u_j (where $i < j$) be two unmarked vertices, such that their horizontal $C(\varphi)$ -full types are the same ($\text{hftp}_{C(\varphi)}^{\mathfrak{T}}(u_i) = \text{hftp}_{C(\varphi)}^{\mathfrak{T}}(u_j)$) and there are no marked vertices between them. Then the tree obtained by removing all vertices between u_i and u_j , including u_i and excluding u_j , together with subtrees rooted at them, is also a model of φ . To prove it, first observe that cutting the vertices between u_i and u_j does not change any of vertices above and below u_i and u_j . The marked vertices guarantee that none of vertices in the whole tree lost its free witnesses. Equality of the horizontal $C(\varphi)$ -full types of u_i and u_j ensures that the numbers of right and left witnesses for vertices u_i and u_j are correct. Therefore the combined $C(\varphi)$ -full type of u_j , realized in the tree after the surgery, is φ -consistent. By Lemma 1.2.10, the obtained tree is a model for the formula φ .



Continuing this process we can remove all vertices between the marked pairs with the same horizontal $C(\varphi)$ -full types. Observe that $\theta_{\rightarrow+}$ and $\theta_{\leftarrow+}$ components of the horizontal $C(\varphi)$ -full types behave in the monotonic way. For fixed $\theta_{\leftarrow}, \theta_{\rightarrow}, \theta_{=}$ components of a given horizontal $C(\varphi)$ -full type, the number of its possible modifications on the path is bounded by $2 \cdot (C(\varphi) + 2) \cdot |\alpha_\varphi|$. This guarantees that between any adjacent marked vertices, we have at most $|\alpha_\varphi|^3 \cdot 2(C(\varphi) + 2) \cdot |\alpha_\varphi|$ vertices. Using the fact that we marked at most $2C(\varphi) \cdot |\alpha_\varphi|$ vertices and we know the upper bound on lengths of paths between adjacent marked vertices, we can reduce the number of children of v to $(4C(\varphi)^2 + 8C(\varphi)) \cdot |\alpha_\varphi|^5$.

By repeating this procedure as long as there are vertices of high degree we obtain a desired model of φ . $\square$

2.2.2 Algorithm

In this section we design an algorithm checking if a given formula $\varphi \in \mathcal{C}^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ has a finite tree model. First, by Lemma 1.2.3, we can assume that φ is in normal form. Second, by Theorem 2.2.1, we can restrict attention to models with exponentially bounded vertex degree and $\downarrow$ -path length.

We will present an alternating algorithm working in exponential space. The idea of the algorithm is quite simple. For each vertex v we will guess its $C(\varphi)$ -full

type and check if it is φ -consistent. If it is, we guess the v 's children and their full types. After that, we check if their $C(\varphi)$ -full types are locally consistent (see the procedure below), which guarantees that we guessed correctly. The algorithm starts with $v = \text{root}$ and works recursively with its children. The procedure presented here is a modification of the one from [2].

Procedure 1 Checking if given $C(\varphi)$ -full types are locally-consistent

Input: $C(\varphi)$ -Full types $\bar{\alpha}, \bar{\alpha}_1, \dots, \bar{\alpha}_k$

- 1: **Return True** if all of the statements below are true. **Return False** otherwise.
 - 2: $\bar{\alpha}_i(\theta_{\leftarrow}) = \bar{\alpha}_{i-1}(\theta_{=})$ for $i > 1$ and $\bar{\alpha}_1(\theta_{\leftarrow}) = \emptyset$
 - 3: $\bar{\alpha}_i(\theta_{\leftarrow+}) = \bar{\alpha}_{i-1}(\theta_{\leftarrow}) \cup \bar{\alpha}_{i-1}(\theta_{\leftarrow+})$ for $i > 1$ and $\bar{\alpha}_1(\theta_{\leftarrow+}) = \emptyset$
 - 4: $\bar{\alpha}_i(\theta_{\rightarrow}) = \bar{\alpha}_{i+1}(\theta_{=})$ for $i < k$ and $\bar{\alpha}_k(\theta_{\rightarrow}) = \emptyset$
 - 5: $\bar{\alpha}_i(\theta_{\rightarrow+}) = \bar{\alpha}_{i+1}(\theta_{\rightarrow}) \cup \bar{\alpha}_{i+1}(\theta_{\rightarrow+})$ for $i < k$ and $\bar{\alpha}_k(\theta_{\rightarrow+}) = \emptyset$
 - 6: $\bar{\alpha}(\theta_{\downarrow}) = \bigcup_{j=1}^k \bar{\alpha}_j(\theta_{=})$
 - 7: $\bar{\alpha}(\theta_{\downarrow+}) = \bigcup_{i=1}^k (\bar{\alpha}_i(\theta_{\downarrow}) \cup \bar{\alpha}_i(\theta_{\downarrow+}))$
 - 8: for $1 \leq i \leq k$: $\bar{\alpha}_i(\theta_{\uparrow}) = \bar{\alpha}(\theta_{=})$
 - 9: for $1 \leq i \leq k$:
 $\bar{\alpha}_i(\theta_{\nearrow}) = \bar{\alpha}(\theta_{\nearrow}) \cup \bar{\alpha}(\theta_{\leftarrow}) \cup \bar{\alpha}(\theta_{\rightarrow}) \cup \bar{\alpha}(\theta_{\leftarrow+}) \cup \bar{\alpha}(\theta_{\rightarrow+}) \cup \bigcup_{j \neq i} (\bar{\alpha}_j(\theta_{\downarrow}) \cup \bar{\alpha}_j(\theta_{\downarrow+}))$
-

Procedure 2 Satisfiability test for $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$

Input: Formula $\varphi \in C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ in normal form.

- 1: Let $\text{MaxDepth} := 3 \cdot (C(\varphi) + 2)^{10m+1} \cdot |\alpha_{\varphi}|^2$
 - 2: Let $\text{MaxDeg} := (4C(\varphi)^2 + 8C(\varphi)) \cdot |\alpha_{\varphi}|^5$
 - 3: $\text{Lvl} := 0$.
 - 4: **guess** a $C(\varphi)$ -full type $\bar{\alpha}$ s.t. $\bar{\alpha}(\theta) = \emptyset$ for $\theta \in \{\theta_{\uparrow}, \theta_{\uparrow+}, \theta_{\rightarrow}, \theta_{\leftarrow}, \theta_{\rightarrow+}, \theta_{\leftarrow+}, \theta_{\nearrow}\}$.
 - 5: **while** $\text{Lvl} < \text{MaxDepth}$ **do**
 - 6: **if** $\bar{\alpha}$ is not φ -consistent **then reject**
 - 7: **if** $\bar{\alpha}(\theta_{\downarrow}) = \bar{\alpha}(\theta_{\downarrow+}) = \emptyset$ **then accept**
 - 8: **guess** an integer $1 \leq k \leq \text{MaxDeg}$
 - 9: **guess** $C(\varphi)$ -full types $\bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_k$
 - 10: **if not** locally-consistent($\bar{\alpha}, \bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_k$) **then reject**
 - 11: $\text{Lvl} := \text{Lvl} + 1$
 - 12: **universally choose** $1 \leq i \leq k$; let $\bar{\alpha} = \bar{\alpha}_i$
 - 13: **end while**
 - 14: **reject**
-

Lemma 2.2.6. Procedure 2 accepts its input φ iff φ is satisfiable.

Proof. Assume φ is satisfiable. Then there exists a small tree model $\mathfrak{T}$ as guaranteed by Theorem 2.2.1. We can run the algorithm and guess exactly the same $C(\varphi)$ -full types as in $\mathfrak{T}$. The guessed $C(\varphi)$ -full types are locally-consistent and φ -consistent, so procedure 2 accepts.

Assume that Procedure 2 accepts its input φ . Then we can reconstruct the tree $\mathfrak{T}$ from the received $C(\varphi)$ -full types. The guessed $C(\varphi)$ -full types are φ -consistent, which guarantees that we have the right number of witnesses to satisfy the formula. Moreover, the function locally-consistent ensures that the $C(\varphi)$ -full types realized in $\mathfrak{T}$ are indeed as we guessed. By Lemma 1.2.10, $\mathfrak{T}$ is a tree model for φ and thus φ is satisfiable.

□

Theorem 2.2.7. The satisfiability problem for $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ over finite trees is EXPSPACE-complete.

Proof. Our procedure works in alternating exponential time, since maximum degree and path length are bounded exponentially in $|\varphi|$. The EXPSPACE-upper bound follows from the well-known fact that AEXPTIME=EXPSPACE. The EXPSPACE-lower bound comes from the previous section.

□

Chapter 3

Conclusions

We proved that $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ remains decidable and retain EXPSPACE-complexity of $FO^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$. There is a lot of open problems in this area:

- It is interesting to investigate the complexity of smaller fragments of $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$. For example we can consider the case when only $\rightarrow$ and $\downarrow$ binary predicates are allowed, or consider unordered trees as a models.
- The satisfiability problem for $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ on infinite trees. Observe that $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ does not have a finite model property – we can consider the formula $\varphi = \forall x \exists y x \downarrow y$, which enforces the existence of infinite paths in models. We believe that methods presented in this paper can be applied in this case.
- Another natural problem to consider the satisfiability problem for $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ on UAR (Unary Alphabet Restriction) trees, that is on trees where for each node exactly one unary predicate is true.
- The succinctness problem. Beside the fact that $FO^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ and $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ share the same complexity and expressive power, we think the second logic allows us to write shorter formulas than the first one. For example the formula "there exists at least n green nodes" seems to require polynomial length in $FO^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$ and only logarithmic length in $C^2[\downarrow, \downarrow^+, \rightarrow, \rightarrow^+]$. Formalizing this result seems to be non-trivial work.

We plan to work on these problems in the nearest future.

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